

12. Cake Mixes An experiment was conducted to compare the densities (in ounces per cubic inch) of cakes prepared from two different cake mixes. Six cake pans were filled with batter A, and six were filled with batter B. Expecting a variation in oven temperature, the experimenter placed a pan filled with batter A and another with batter B *side by side* at six different locations in the oven. The six paired observations of densities are as follows:

Location	1	2	3	4	5	6
Batter A	.135	.102	.098	.141	.131	.144
Batter B	.129	.120	.112	.152	.135	.163
	0.006	-0.018	-0.014	-0.011	-0.004	-0.019

- Do the data present sufficient evidence to indicate a difference between the average densities of cakes prepared using the two types of batter?
- Construct a 95% confidence interval for the difference between the average densities for the two mixes.

令 X_1 : Batter A, X_2 : Batter

$\bar{D} = -0.01$, $S_D \approx 0.00953$

a. $H_0: D_0 = 0$, $H_A: D_0 \neq 0$, $\alpha = 0.05$

$t_{STAT} = \frac{\bar{D} - D_0}{\frac{S_D}{\sqrt{n}}} \sim t_5$

$t^* = \frac{-0.01 - 0}{\frac{0.00953}{\sqrt{6}}} = -2.570$

rejection region: $|t| > 2.571$

$t_{5; 0.025} = 2.571$

$|t^*| = |-2.570| < 2.571 \Rightarrow \text{not reject } H_0$

b. $CI = \bar{D} \pm t_{5; 0.025} \cdot \frac{S_D}{\sqrt{n}}$
 $= -0.01 \pm 2.571 \cdot \frac{0.00953}{\sqrt{6}}$
 $= (-0.020, 0.0000027)$